به نام خدا

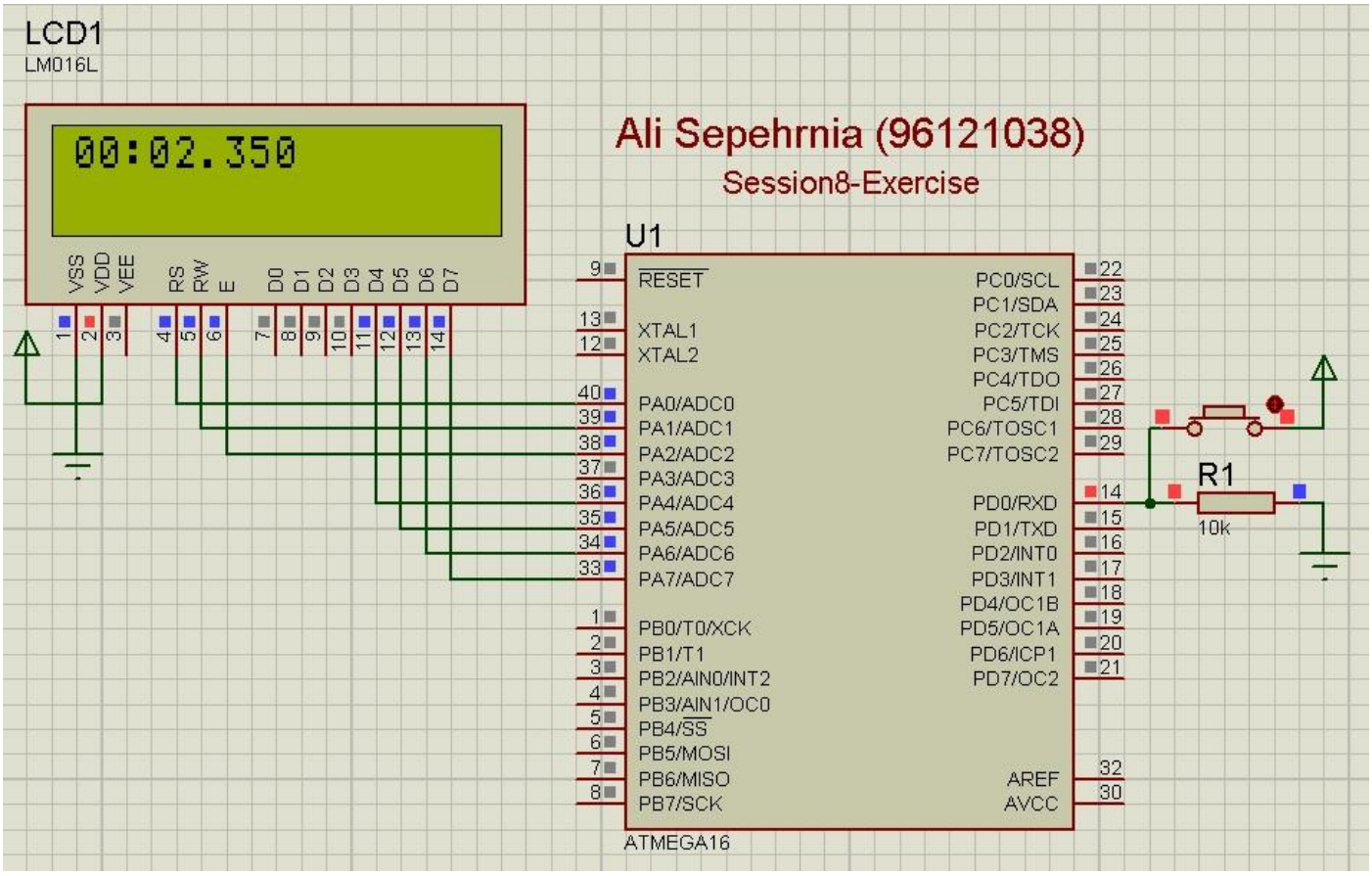
علی سپهرنیا (۹۶۱۲۱۰۳۸)

آزمایشگاه سیستم دیجیتال ۲

استاد: خانم رضایی

پاسخ تمرین جلسه هشتم

برنامه ای بنویسید که مدت زمانی که کلید فشاری فشرده نگه داشته میشود را بر حسب میلی ثانیه به دست آورده و بر روی lcd کاراکتری بنویسد. (از مد نرمال تایمر ۱ استفاده کنید)



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Notes | Az8-Tamrin.c

```
/*****************************************************
This program was created by the
CodeWizardAVR V3.12 Advanced
Automatic Program Generator
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Project : Session8-Exercise
Version :
Date    : 11/20/2020
Author  : Ali Sepehrnia(96121038)
Company :
Comments:


Chip type               : ATmega16
Program type            : Application
AVR Core Clock frequency: 1.000000 MHz
Memory model            : Small
External RAM size       : 0
Data Stack size         : 256
*****************************************************/

#include <mega16.h>
#include <delay.h>
#include <stdlib.h>
#include <stdio.h>

// Alphanumeric LCD functions
#include <alcd.h>

// Declare your global variables here
unsigned char ms=0; //Based on 1ms (For Debouncing)
unsigned int time=0; //Based on 50ms (Can store up to 54 Minutes!)

```

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```
// Timer1 overflow interrupt service routine
interrupt [TIM1_OVF] void timer1_ovf_isr(void)
{
// Reinitialize Timer1 value
TCNT1H=0xFC18 >> 8;
TCNT1L=0xFC18 & 0xff;

    //Add 1ms To Timer If Key Pressed
    if(PIND.0==1)
    {
        ms++;
    }
    //Reset Timers If Key Unpressed
    else
    {
        ms=0;
        time=0;
    }
    //Reset Bounce Timer If Button Pressed For More Than 50ms (Debounce)
    if(ms>50)
    {
        ms=0;
        time++;
    }
}

void main(void)
{
// Declare your local variables here

// Input/Output Ports initialization
// Port A initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRA=(0<<DDA7) | (0<<DDA6) | (0<<DDA5) | (0<<DDA4) | (0<<DDA3) | (0<<DDA2) | (0<<DDA1) | (0<<DDA0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTA=(0<<PORTA7) | (0<<PORTA6) | (0<<PORTA5) | (0<<PORTA4) | (0<<PORTA3) | (0<<PORTA2) | (0<<PORTA1) | (0<<PORTA0);
```

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```
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRB=(0<<DDB7) | (0<<DDB6) | (0<<DDB5) | (0<<DDB4) | (0<<DDB3) | (0<<DDB2) | (0<<DDB1) | (0<<DDB0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTB=(0<<PORTB7) | (0<<PORTB6) | (0<<PORTB5) | (0<<PORTB4) | (0<<PORTB3) | (0<<PORTB2) | (0<<PORTB1) | (0<<PORTB0);

// Port C initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRC=(0<<DDC7) | (0<<DDC6) | (0<<DDC5) | (0<<DDC4) | (0<<DDC3) | (0<<DDC2) | (0<<DDC1) | (0<<DDC0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTC=(0<<PORTC7) | (0<<PORTC6) | (0<<PORTC5) | (0<<PORTC4) | (0<<PORTC3) | (0<<PORTC2) | (0<<PORTC1) | (0<<PORTC0);

// Port D initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRD=(0<<DDD7) | (0<<DDD6) | (0<<DDD5) | (0<<DDD4) | (0<<DDD3) | (0<<DDD2) | (0<<DDD1) | (0<<DDD0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTD=(0<<PORTD7) | (0<<PORTD6) | (0<<PORTD5) | (0<<PORTD4) | (0<<PORTD3) | (0<<PORTD2) | (0<<PORTD1) | (0<<PORTD0);

// Timer/Counter 1 initialization
// Clock source: System Clock
// Clock value: 1000.000 kHz
// Mode: Normal top=0xFFFF
// OC1A output: Disconnected
// OC1B output: Disconnected
// Noise Canceler: Off
// Input Capture on Rising Edge
// Timer Period: 1 ms
// Timer1 Overflow Interrupt: On
// Input Capture Interrupt: Off
// Compare A Match Interrupt: Off
// Compare B Match Interrupt: Off
TCCR1A=(0<<COM1A1) | (0<<COM1A0) | (0<<COM1B1) | (0<<COM1B0) | (0<<WGM11) | (0<<WGM10);
TCCR1B=(0<<ICNC1) | (1<<ICES1) | (0<<WGM13) | (0<<WGM12) | (0<<CS12) | (0<<CS11) | (1<<CS10);
TCNT1H=0xFC;
TCNT1L=0x18;
ICR1H=0x00;
```

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```
ICR1L=0x00;
OCR1AH=0x00;
OCR1AL=0x00;
OCR1BH=0x00;
OCR1BL=0x00;

// Timer(s)/Counter(s) Interrupt(s) initialization
TIMSK=(0<<OCIE2) | (0<<TOIE2) | (0<<TICIE1) | (0<<OCIE1A) | (0<<OCIE1B) | (1<<TOIE1) | (0<<OCIE0) | (0<<TOIE0);

// Alphanumeric LCD initialization
// Connections are specified in the
// Project|Configure|C Compiler|Libraries|Alphanumeric LCD menu:
// RS - PORTA Bit 0
// RD - PORTA Bit 1
// EN - PORTA Bit 2
// D4 - PORTA Bit 4
// D5 - PORTA Bit 5
// D6 - PORTA Bit 6
// D7 - PORTA Bit 7
// Characters/line: 16
lcd_init(16);

// Global enable interrupts
#asm("sei")

while (1)
      {
        unsigned long milliseconds = time*50; //Convert Time To Milliseconds
        unsigned char minutes,seconds;
        char str[10];

        seconds = milliseconds / 1000;
        milliseconds %= 1000;

        minutes = seconds / 60;
        seconds %= 60;
```

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```
        seconds %= 60;

        sprintf(str,"%02u:%02u.%03u", minutes, seconds, milliseconds);
        lcd_clear();
        lcd_puts(str);

        delay_ms(50);
      }
}
```